CPSC 481
Artificial Intelligence

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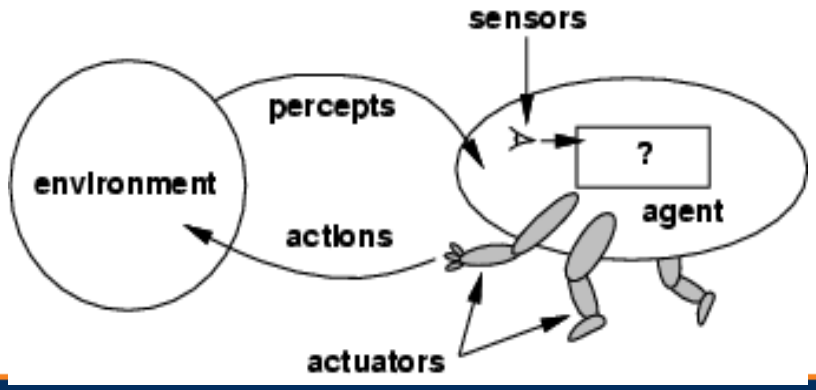
What we will cover today

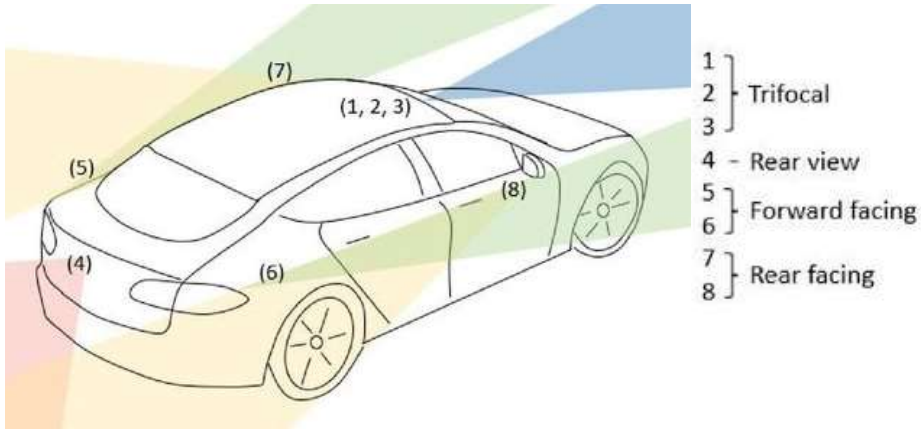
- Types of environments

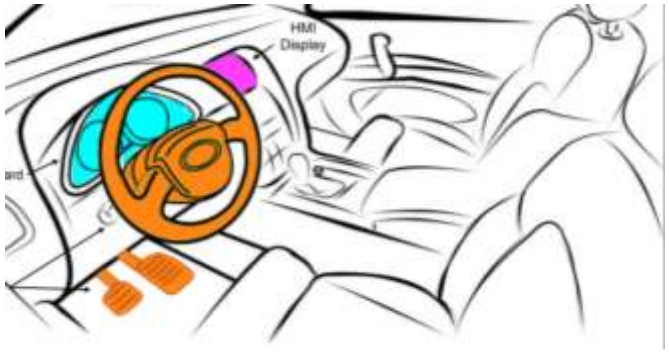
Intelligent Agents

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators
  - Human agent has eyes, ears, and other organs for sensors; hands, legs, mouth, and other body parts for actuators.
  - Robotic agent has cameras and infrared range finders for sensors; various motors for actuators.
- The agent function maps from percept histories to actions: $[f: \mathcal{P}^* \rightarrow \mathcal{A}]$
- The agent program runs on the physical architecture to produce
  - agent = architecture + program
- Rational agents use
  - performance measures
  - environment
  - actuators
  - sensors



Self-driving car sensors and actuators





Mandujano-Granillo, Jesus & Candela, Milton & Ortiz-Vazquez, Juan & Ramírez Moreno, Mauricio Adolfo & Tudón-Martínez, Juan & Félix-Herrán, Luis Carlos & Galvan-Galvan, Alfredo & Lozoya-Santos, Jorge. (2024). Human-Machine Interfaces: A Review for Autonomous Electric Vehicles. IEEE Access. 12. 121635–121658. 10.1109/ACCESS.2024.3450439.

Class work

- List the
  - performance measures
  - environment
  - actuators
  - sensors
- For (each of):
  - A self-driving wheelchair
  - A robot vacuum cleaner (Roomba)
  - A robot soccer player

Classwork

- On Google Drive
  - **Link on Canvas**
- Go to today's date and section
  - Editable and viewable by all
- Create a new Google Document and name it with your groupmates' names
  - E.g.: Garcia-Joshi-Nguyen
  - Write the names and role of group members
  - Roles: manager, recorder+presenter, timekeeper
  - Add your answers
- Will not be graded but used for class participation points
  - Can easily share with rest of class

Properties of Task Environments

- **Fully observable vs. partially observable**: – Fully observable if sensors detect all aspects of environment relevant to choice of action
  - Partially observable due to noisy, inaccurate or missing sensors
  - E.g., in card games, not possible to see what cards others are holding
- **Single agent vs. multiple agents: in single agent environment, no other "thinking" entity**
  - Multiple agents: other entities whose goals depend on your own
  - Cooperative: improving other's situation, helps your own
  - Competitive: improving other's situation, worsens your own
- **Deterministic vs. stochastic**: Deterministic if the next state of the environment is completely determined by the current state and the action executed by the agent
  - Stochastic if what happens is affected by randomness
  - E.g., roll of dice in a game

Properties of Task Environments-cont.

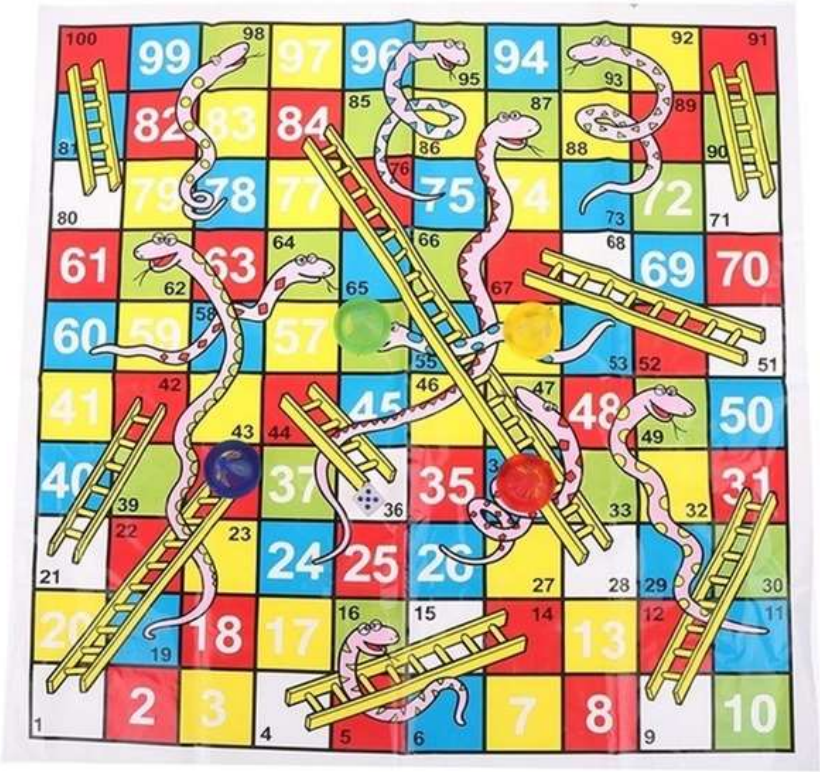
- **Episodic vs. sequential:** episodic if a current action will not affect future actions - the agent's experience is divided into independent episodes. E.g., mail sorting system
  - Sequential if current decisions affect future decisions.
  - E.g., navigating a car is sequential, most games
- **Static vs. dynamic**: Static if the environment does not change as the agent is deciding on an action. E.g., chess
  - dynamic if the environment may change while the agent is thinking
- **Discrete vs. continuous**: in discrete environments, actions/percepts can take only specific values (e.g., chess); in continuous environments, there can be infinite such values.
  - Signals constantly coming into sensors, actions continually changing. Car driving is continuous.
- **Known vs. unknown**: In a known environment, the outcomes for all actions are given, i.e., the rules of the game are known
  - If unknown, the agent will have to learn how it works.

Example: playing soccer

- Partially observable – Player cannot detect all the things on soccer field that can affect its action, for e.g. it cannot determine what other players are thinking.
- Multi-agent – There are many players in a soccer game.
- Stochastic – Given the current state and action executed by agent, the outcome cannot be exactly determined. E.g., if player kicks the ball, then the ball may or may not be stopped by other players, or the soccer field can change in many different ways depending on how players move.
- Sequential – The past history of actions in the game can affect the next action in the game.
- Dynamic – The environment can change while the agent is making decision, for e.g., position of other players changes when a player moves.
- Continuous – Positions of the ball and players are continuous. The speed or the direction (angle) at which the player kicks the ball is continuous.

Chutes and ladders

- Fully observable vs. partially observable?
- Deterministic vs. stochastic?
- Episodic vs. sequential?
- Static vs. dynamic?
- Discrete vs. continuous?
- Known vs. unknown?
- Single agent vs. multiple agents?



Passing a CAPTCHA

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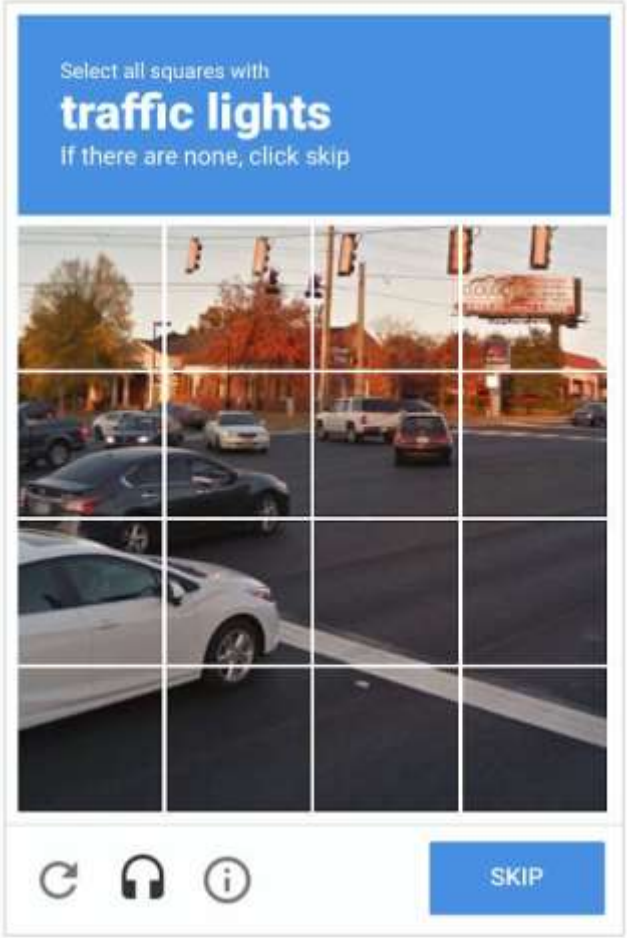

Chess

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Driving a car

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- Episodic vs. sequential?
- Static vs. dynamic?
- Discrete vs. continuous?
- Known vs. unknown?
- Single agent vs. multiple agents?

Class work

- List the properties for each of the previous tasks
  - Chutes and ladders
  - Passing a CAPTCHA
  - Chess
  - Driving a car

References

- Russel and Norvig, Artificial Intelligence: A Modern Approach, 3rd edition, Prentice Hall, 2010. **Chapter 3.3**